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Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date: 31/07/2026

Project Category:	☒ Research oriented	☐ Application oriented	Batch ID:	AIML-B17
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Scaling Direct Cloud Upload Auditing via Autonomous AI Agents

Abstract

Modern web applications increasingly use direct cloud upload architectures, where users upload files directly to cloud storage services such as Amazon S3, Google Cloud Storage, Microsoft Azure Blob Storage, and Alibaba Cloud OSS. This approach reduces server workload, improves scalability, and enhances upload performance. However, it also introduces security risks such as unauthorized upload credential access, credential validity flaws, unrestricted file uploads, file overwriting, file stealing, and callback notification spoofing. These vulnerabilities can lead to data leakage, resource abuse, and financial losses for organizations using cloud-based storage services. Existing research has systematically identified these vulnerabilities and proposed mitigation techniques, but the security evaluation process still depends largely on manual and semi-automated testing. As a result, assessing a large number of web applications becomes time-consuming and difficult, limiting the scalability of security auditing. This project, **Scaling Direct Cloud Upload Auditing via Autonomous AI Agents**, proposes an intelligent AI-based security auditing framework to automate vulnerability detection in direct cloud upload systems. The proposed framework employs autonomous AI agents to monitor upload workflows, analyse upload credentials, verify access control policies, inspect file upload restrictions, and detect security weaknesses through automated testing. Compared with traditional manual approaches, the framework can audit a larger number of web applications with minimal human intervention, improving both efficiency and coverage. The project is limited to security assessment in controlled testing environments and does not involve attacks on production systems. The expected outcome is a scalable and automated security auditing solution that enables faster, more accurate vulnerability detection and assists developers in strengthening the security of cloud-based web applications.

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References (Category specific)

1. Y. Chen et al., "Understanding the Security Risks of Websites Using Cloud Storage for Direct User File Uploads," *IEEE TIFS*, 2025.
• <https://ieeexplore.ieee.org/document/10896886>
2. G. Hong et al., "Understanding and Detecting Abused Image Hosting Modules as Malicious Services," *ACM CCS*, 2023.
• <https://ghong.site/papers/ccs23b-imagehosting.pdf>
3. Y. Lv et al., "Don't Trust the Clouds Easily: The Insecurity of Content Security Policy Based on Object Storage," *IEEE IoT Journal*, 2023.
• <https://ieeexplore.ieee.org/document/10022315>
4. E. Trickel et al., "Toss a Fault to Your Witcher: Applying Grey-Box Coverage-Guided Mutational Fuzzing," *IEEE S&P*, 2023.
• <https://ieeexplore.ieee.org/document/10179317>
5. Amazon Web Services, "Amazon Simple Storage Service (Amazon S3) Documentation."
• <https://docs.aws.amazon.com/s3/>
6. I. David & A. Gervais, "Multi-Agent Penetration Testing AI for the Web," *arXiv*, 2025.
• <https://arxiv.org/abs/2508.20816>
7. Microsoft Azure, "Azure Blob Storage Documentation."
• <https://learn.microsoft.com/en-us/azure/storage/blobs/>
8. OWASP Foundation, "OWASP Top 10 Web Application Security Risks," 2021.
• <https://owasp.org/Top10/2025/>
9. NIST, "Security and Privacy Controls for Information Systems," SP 800-53 Rev. 5, 2020.
• <https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final>
10. Cloud Security Alliance, "Security Guidance for Critical Areas of Cloud Computing," v5, 2024.
• <https://cloudsecurityalliance.org/research/guidance/>

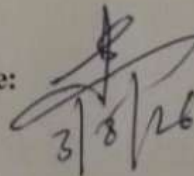
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